HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented automation solutions for manufacturing equipment, including PLC programming and system integration, enhancing operational reliability.
- Led automation project activities for equipment upgrades and process improvements, ensuring successful integration and validation.
- Authored functional specifications, design documentation, and test protocols for automation systems, ensuring compliance with GMP and regulatory requirements.
- Provided technical coaching and support to Automation Technicians, enhancing team capabilities in troubleshooting and system maintenance.
- Analyzed manufacturing process performance data to identify opportunities for automation enhancement and efficiency gains, contributing to plant performance improvements.
- Managed lifecycle of automation hardware and software platforms, ensuring systems remained up-to-date and compliant with company standards.
- Collaborated with equipment vendors and cross-functional teams during project execution, ensuring seamless integration of new automation systems.

SKILLS

PLC/Control Systems – Allen Bradley ControlLogix, CompactLogix

Machine Vision – Cognex Vision System

Networking/Integration – Ethernet/IP, Industrial Networks, HMIs, SCADA, MES

Software/Platforms – Rockwell Automation Software, Manufacturing Control Architectures

Methodologies – Root Cause Analysis, Problem-Solving, Change Management, Validation Methodologies, Lean Manufacturing

Project Management – Project Management, Planning, Cross-functional Communication

PROJECTS

Manufacturing Automation System Enhancement

- Designed and implemented an automated quality inspection system using **Cognex Vision System** applications for a key production line, reducing manual inspection errors by **28%**.
- Developed new logic and modified existing programs for **Allen Bradley ControlLogix** PLCs to integrate a new robotic arm for material handling, improving throughput by **15%**.
- Created detailed test protocols and executed validation procedures for the new automation features, ensuring adherence to **GMP requirements** and company standards.

Production Data Collection and Analysis Platform

- Engineered an automation interface to collect real-time performance data from manufacturing equipment, integrating with **SCADA systems** and **MES interfaces**.
- Developed Python scripts to parse and analyze collected data, identifying bottlenecks and providing actionable insights for process improvements to operations teams.
- Designed and implemented database schemas in **PostgreSQL** to store historical production data, enabling detailed trend analysis and reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)